DO NOW, AO1 REVISION

Mass number, atomic number and isotopes

Atomic structure
Paper 1, Combined and Separate
11 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**State the overall charge of an atom.**	Zero	☐
2	**Explain why atoms have an overall neutral charge.**	The number of protons equals the number of electrons.	☐
3	**State what is the same about all atoms of the same element.**	The number of protons	☐
4	**State what the atomic number of an atom tells you.**	The number of protons	☐
5 ★	**State what the mass number of an atom tells you.**	The total number of protons and neutrons	☐
6 ★	**Define isotopes.**	Atoms of the same element with different numbers of neutrons.	☐
7	**Which differs between two isotopes of an element: mass number or atomic number?**	The mass number	☐
8	**Describe how ions form from atoms.**	Atoms lose or gain electrons.	☐
9	**An atom of carbon is written $^{14}_{6}C$. State what the numbers 14 and 6 represent.**	14: the mass number (protons + neutrons). 6: the atomic number (protons).	☐
10 ★	**State how to calculate the number of neutrons in an atom.**	Number of neutrons = mass number − atomic number	☐
11 ★	**State the type of ion formed when an atom loses one or more electrons.**	A positive ion	☐

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